

Session 4 Lab - Multivariate OLS

Women in Politics and Military Spending

This lab uses the `women12.dta` dataset from Koch & Fulton (2011) examining the relationship between women's political representation and national security policy in 22 established democracies (1970-2000). You'll practice multivariate regression, dummy variable interpretation, and model comparison.

Research Question: Do women's political gains translate into substantive differences in military spending?

Setup

```
# Load required libraries
library(haven) # for reading .dta files
library(ggplot2)

# Load the data
women12 <- read_dta("women12.dta")

# Create variables we need for analysis
women12$leftright <- women12$DPI_Left # Left government dummy (0/1)
women12$partycontrol <- women12$partcont # Party control scale (0-2)

# Examine the data structure
str(women12)
head(women12)
```

Part 1: Data Exploration

1.1 Explore the key variables:

- `mil_exp_per_gdp2`: Military expenditure as % of GDP (dependent variable)
- `m_woman2`: Percentage of women in parliament
- `m_femlead`: Binary variable (1 = female leader, 0 = male leader)
- `leftright`: Government ideology (created from `DPI_Left`: 1 = left government, 0 = non-left)
- `partycontrol`: Degree of party control over legislators (created from `partcont`: 0-2 scale)

1.2 Create summary statistics and visualizations:

- What's the range of military spending across countries?
- How much variation is there in women's parliamentary representation?

- How many female leaders are in the dataset?

1.3 Create a scatter plot of women in parliament vs military spending. What initial relationship do you observe?

Part 2: Simple vs Multivariate Comparison

2.1 Fit a simple bivariate model:

```
# Simple model: Military spending ~ Women in parliament
simple_model <- lm(mil_exp_percgdp2 ~ m_woman2, data = women12)
```

2.2 Fit a multivariate model controlling for government ideology:

```
# Multivariate model: Add government left-right ideology (dummy variable)
multi_model <- lm(mil_exp_percgdp2 ~ m_woman2 + leftright, data = women12)
```

2.3 Compare the coefficients:

- How does the coefficient on `m_woman2` change between models?
- Why might controlling for government ideology (left vs non-left) matter?
- Which model has higher R-squared?

Part 3: Understanding Dummy Variables

3.1 Examine the female leader variable:

- Create a table showing the distribution of `m_femlead`
- What percentage of observations have female leaders?

3.2 Add the female leader dummy to your model:

```
# Model with both women in parliament and female leader dummy
dummy_model <- lm(mil_exp_percgdp2 ~ m_woman2 + m_femlead + leftright, data = women12)
```

3.3 Interpret the dummy variable coefficient:

- What does the `m_femlead` coefficient tell you?
- How do you interpret this in substantive terms?
- Is the effect statistically significant?

Note: Remember that `leftright` is also a dummy variable (1 = left government, 0 = non-left government).

Part 4: Creating Your Own Dummy Variables

4.1 Create a dummy variable for high women's representation:

```
# Create dummy: 1 if women's representation > median, 0 otherwise
women12$high_women <- ifelse(women12$m_woman2 > median(women12$m_woman2,
na.rm = TRUE), 1, 0)
```

4.2 Create a categorical variable for party control:

```
# Create three categories of party control from the 0-2 scale
women12$party_control_cat <- cut(women12$partycontrol,
                                breaks = c(-0.1, 0.5, 1.5, 2.1),
                                labels = c("Low", "Medium", "High"),
                                include.lowest = TRUE)
```

4.3 Run models with your new dummy variables:

- Test the effect of high women's representation (binary) vs continuous representation
- Test the effect of different levels of party control

Part 5: Model Comparison and Interpretation

5.1 Fit a comprehensive model:

```
# Full model with multiple controls
full_model <- lm(mil_exp_percdp2 ~ m_woman2 + m_femlead + leftright + party_control_cat,
                data = women12)
```

5.2 Compare all your models:

- Create a table comparing R-squared values
- Which variables remain significant across specifications?
- How do the women's representation effects change?

5.3 Interpret each coefficient in the full model:

- What does each coefficient mean in practical terms?
- Which effects are substantively large vs small?

Part 6: Predictions and Scenarios

6.1 Make predictions for different scenarios:

- Predict military spending for a country with 10% women in parliament, male leader, non-left government
- Predict for a country with 40% women in parliament, female leader, left government
- What's the predicted difference between these scenarios?

6.2 Test the effect of changing leadership gender:

- For a country with 25% women in parliament and non-left government, what's the predicted difference between male and female leadership?

6.3 Examine party control effects:

- Holding other variables constant, how does military spending differ across low, medium, and high party control systems?

Part 7: Testing the Koch & Fulton Hypotheses

Based on the paper, test these key hypotheses:

7.1 Legislative hypothesis:

"As the proportion of women in the legislature increases, defense spending decreases"

- Is this supported in your models?
- How large is the effect?

7.2 Executive hypothesis:

"If the chief executive is a woman, defense spending increases"

- Does your female leader variable support this?
- How does this relate to gender stereotypes in politics?

7.3 Institutional context:

- Do the effects of women's representation vary by party control?
- Create an interaction between `m_woman2` and `party_control_cat`

Part 8: Advanced Analysis

8.1 Create interaction terms:

```
# Test if female leaders behave differently when more women are in parliament
interaction_model <- lm(mil_exp_per_gdp2 ~ m_woman2 + m_femlead +
  m_woman2:m_femlead + leftright,
  data = women12)
```

8.2 Interpret the interaction:

- What does the interaction coefficient tell you?
- Plot the predicted values for male vs female leaders across different levels of women's representation

8.3 Test for robustness:

- Do your results hold if you exclude potential outliers?
- How sensitive are your results to model specification?

Part 9: Visualization

9.1 Create enhanced plots:

- Scatter plot with different colors for male vs female leaders
- Box plots showing military spending by party control categories
- Plot showing predicted values from your interaction model

9.2 Create a coefficient plot:

- Show coefficients and confidence intervals from your main models
- Which effects are most precisely estimated?

Reflection Questions

1. **Theory vs Evidence:** Do your results support the theoretical predictions about gender and military spending? Why or why not?
2. **Omitted Variables:** What important variables might be missing from these models? How might their omission bias your results?
3. **Causal Interpretation:** Can you make causal claims about the effect of women's representation on military spending? What are the limitations?
4. **Policy Implications:** If these relationships are causal, what would they suggest for those interested in reducing military spending?
5. **Institutional Context:** How do electoral institutions (party control) seem to matter for women's influence on policy?
6. **Gender Stereotypes:** How do your results relate to theories about gender stereotypes in politics? Do female executives really behave more "hawkishly"?

Important: We've created two key variables from the original data:

- `leftright` from `DPI_Left` (dummy for left government)
- `partycontrol` from `partcont` (0-2 scale for party control)